

Module 5

1. Key Features of Wi-Fi 6, 6E, and 7 vs Wi-Fi 5

Wi-Fi 5 (802.11ac)

Wi-Fi 5 mainly focused on increasing speed in the 5 GHz band using:

- OFDM
- MU-MIMO (downlink mainly)
- 80/160 MHz channels
- 256-QAM

It improved throughput but struggled in crowded environments.

Wi-Fi 6 (802.11ax)

Wi-Fi 6 improves efficiency and performance in dense networks.

Main features:

- OFDMA
- Uplink + Downlink MU-MIMO
- Target Wake Time (TWT)
- BSS Coloring
- 1024-QAM

Benefits:

- Better battery life
- Lower latency
- Higher efficiency

- Better performance in crowded areas

Wi-Fi 6E

Wi-Fi 6E adds support for the new **6 GHz band**.

Benefits:

- More clean spectrum
- Less interference
- More wide channels
- Better performance for high-speed applications

It uses the same technologies as Wi-Fi 6 but operates in 6 GHz also.

Wi-Fi 7 (802.11be)

Wi-Fi 7 introduces Extremely High Throughput (EHT).

Main features:

- Multi-Link Operation (MLO)
- 320 MHz channels
- 4096-QAM
- Enhanced OFDMA
- Multi-RU allocation

Benefits:

- Very low latency
- Extremely high throughput
- Better VR/AR/gaming performance

2. Role of OFDMA in Wi-Fi 6

OFDMA stands for Orthogonal Frequency Division Multiple Access.

In older OFDM systems:

- One user occupies the entire channel at a time.

In OFDMA:

- The channel is divided into smaller Resource Units (RUs).
- Multiple users transmit simultaneously.

Benefits:

- Lower latency
- Better efficiency
- Reduced contention
- Improved performance in crowded networks

It is especially useful for:

- IoT devices
- Dense enterprise environments

3. Benefits of Target Wake Time (TWT)

TWT allows devices to schedule exactly when they wake up to communicate.

Instead of constantly listening:

- Devices sleep most of the time
- Wake only at scheduled intervals

Benefits:

- Huge battery savings
- Reduced channel contention
- Better IoT scalability

Very useful for:

- Sensors
- Smart home devices
- Battery-powered IoT devices

4. Significance of the 6 GHz Band in Wi-Fi 6E

The 6 GHz band provides a large amount of new clean spectrum.

Advantages:

- Less congestion
- Less interference
- More non-overlapping channels
- More 160 MHz channels

This improves:

- Throughput
- Latency
- Reliability

It is ideal for:

- 4K/8K streaming
- AR/VR
- Gaming
- High-density enterprise networks

5. Wi-Fi 6 vs Wi-Fi 6E

Feature	Wi-Fi 6	Wi-Fi 6E
Bands	2.4 & 5 GHz	2.4, 5 & 6 GHz
Interference	Moderate	Very low in 6 GHz
Range	Better	Slightly shorter at 6 GHz
Channel Availability	Limited	Much larger
Congestion	Higher	Lower

6 GHz provides cleaner spectrum but slightly lower range because higher frequencies attenuate faster.

6. Major Innovations in Wi-Fi 7

Wi-Fi 7 introduces several major improvements:

Multi-Link Operation (MLO)

Allows simultaneous use of multiple bands/channels.

320 MHz Channels

Doubles channel width from Wi-Fi 6.

4096-QAM

Higher modulation for greater throughput.

Multi-RU

Improves OFDMA flexibility.

Lower Latency

Important for gaming and AR/VR.

Wi-Fi 7 is designed for:

- Real-time applications
- Ultra-high throughput
- Deterministic low latency

7. Multi-Link Operation (MLO)

MLO allows a device to use multiple Wi-Fi links simultaneously.

Example:

- Device uses 5 GHz + 6 GHz together.

Benefits:

- Higher throughput
- Lower latency
- Better reliability
- Faster failover

If one link becomes congested, traffic can move to another link immediately.

This is one of the most important Wi-Fi 7 features.

8. Purpose of 802.11k and 802.11v in Roaming

802.11k — Neighbor Reports

Helps clients know nearby APs.

The AP provides:

- Neighbor AP list
- Channel information
- Signal details

This reduces scanning time during roaming.

802.11v — Network Assisted Roaming

Allows the network to guide clients toward better APs.

Benefits:

- Load balancing
- Better roaming decisions
- Improved performance

9. Fast BSS Transition (802.11r)

802.11r improves fast roaming between APs.

Normally during roaming:

- Full authentication takes time.

802.11r allows:

- Pre-generated security keys
- Faster handoff

Benefits:

- Reduced roaming delay
- Better VoIP/video performance
- Seamless mobility

Very important in:

- Hospitals
- Warehouses
- Enterprise Wi-Fi

10. How 802.11k/v/r Work Together

These standards together provide seamless enterprise roaming.

802.11k

Tells client which APs are nearby.

802.11v

Suggests the best AP to move to.

802.11r

Performs very fast secure handoff.

Together they:

- Reduce roaming delay
- Improve voice/video quality
- Minimize packet loss
- Improve user experience while moving

This combination is widely used in enterprise WLAN deployments.